

Gapped Kernel Realization in Clique Filtrations: Exact Multiplicity and True Normalized Persistence

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Abstract

For an inclusion of clique complexes $X_1 \subseteq X_2$, true normalized persistence is the fraction

$$\frac{\text{rank}[H_d(X_1) \rightarrow H_d(X_2)]}{\dim H_d(X_1)}.$$

Hardness reductions for this quantity must control three features simultaneously: exact homology multiplicities, the inclusion-induced map, and an inverse-polynomial gap above the complete kernel of each endpoint Laplacian. We give a finite-certificate transfer theorem that supplies all three for a fixed palette of weighted clique gadgets. Its main estimate applies to every geometric chain and bounds squared distance to an embedded degenerate logical kernel by

$$C \left(t\lambda^2 + \frac{\langle x, \Delta x \rangle}{g\lambda^{26}} \right).$$

The gadget weight λ is independent of the logical gap g , yielding a geometric gap linear in g . Exact filling identifies the endpoint homology as a quotient V/W_A ; nested term sets induce the natural quotient epimorphisms, so persistent rank equals the later logical-kernel dimension. Applying the theorem to an exact real-gate perfect-completeness verifier gives weighted instances with $\beta_d(X_1) = 8$, persistent rank 6 in YES instances and 1 in NO instances, and inverse-polynomial whole-kernel gaps. Thus additive true normalized persistence is hard for the gate-dependent class $\text{BQP}_1^{G_2}$, where $G_2 = \{X, CX, CCX, H \otimes H\}$. A common-copy construction transfers the result to unweighted clique filtrations. The theorem does not imply ordinary BQP, unrestricted SDQC_1 , or gate-independent BQP_1 hardness.

1 Introduction

Persistent homology asks which topological features survive as a simplicial complex grows. For a two-step filtration $X_1 \subseteq X_2$, the basic numerical invariant is the rank of the induced map $H_d(X_1) \rightarrow H_d(X_2)$. Dividing this rank by $\beta_d(X_1)$ gives an interpretable fraction: among the degree- d classes present initially, what proportion survives? We call this the *true normalized persistence ratio*, following the formulation of Lowe, Kim, Bondesan, and Hayakawa [5].

This ratio sits between several nearby problems that have different mathematical requirements. Exact clique homology can encode the multiplicity of a quantum satisfying space parsimoniously [1], yet the resulting Laplacian need not have an inverse-polynomial gap above all of its zero modes. Low-energy counting permits small positive eigenvalues and therefore does not determine an exact Betti number without a gap promise. Normalization by the number of simplices measures a different density. Finally, following one specified harmonic guide through a filtration [2] does not reveal the fraction of an entire initial homology group that survives. A reduction for true normalized persistence must close all of these gaps at once.

The technical obstruction is degeneracy. Existing clique-gadget analyses establish a spectral gap when a simulated logical Hamiltonian is positive definite [4]. A persistence reduction instead needs two logical Hamiltonians with known, generally nonzero kernels. It is insufficient to construct the desired zero modes. One must rule out every additional geometric zero mode and every positive eigenvalue below the promised threshold, including vectors supported largely outside the simulated register.

We address this obstruction through a finite-certificate all-chain theorem. The theorem assumes a fixed palette of clique gadgets. Each gadget supplies exact filling data, a zero-weight kernel decomposition, coercivity of a projected private differential, a local positive spectral floor, and a bounded interface with the common register. Unused register factors are padded by their harmonic spaces in every chain bidegree. These conditions are finite and can be checked exactly for a fixed palette. They imply, for every unit geometric chain x ,

$$\text{dist}(x, K_A)^2 \leq C \left[t_A \lambda^2 + \frac{\langle x, \Delta_A x \rangle}{g_A \lambda^\kappa} \right], \quad \kappa \leq 26, \quad (1)$$

where K_A is the embedded kernel of the logical projector sum H_A , g_A is its smallest positive eigenvalue, and Δ_A is the full geometric Laplacian. The estimate concerns arbitrary geometric vectors; it is not a statement about distance to the geometric Laplacian's own kernel.

The form of equation (1) matters quantitatively. A direct use of the earlier many-gadget master inequality would choose λ proportional to g_A , producing a high power of the logical gap in the final estimate. Here the structural error is $t_A \lambda^2$, while the energy term is divided by $g_A \lambda^\kappa$. We may choose λ from the palette size and target angle, independently of g_A . For locality at most six, a conservative choice gives

$$E_A = \Omega\left(\frac{\eta^{28} g_A}{t_A^{26}}\right).$$

The exponent is poor, but the scale is inverse polynomial and linear in the logical gap.

Exact filling then turns the analytic estimate into an exact topological statement. Let $V = Z_d(R)$ be the cycle space of the top-dimensional register and let W_A be the span of the logical projector ranges attached by the term set A . Independent gadget interiors give

$$B_d(X_A) \cap V = W_A.$$

This supplies an injection $V/W_A \hookrightarrow H_d(X_A)$. Equation (1) bounds the whole low-energy spectral subspace in the reverse direction, yielding

$$H_d(X_A) \cong V/W_A, \quad \dim \ker \Delta_A = \dim \ker H_A, \quad \text{spec}(\Delta_A) \cap (0, E_A) = \emptyset.$$

For $A \subseteq B$, simplicial inclusion becomes the quotient epimorphism $V/W_A \twoheadrightarrow V/W_B$. The persistent rank is therefore exactly $\dim \ker H_B$. This quotient description avoids any need to choose harmonic representatives compatibly across the filtration.

Our complexity application uses an exact perfect-completeness circuit source. A three-bit mixed label transforms a G_2 -verifier with acceptance probability p_x into the compressed operator

$$M_x = \text{diag}(1, p_x, p_x, p_x, p_x, p_x, 0, 0).$$

In a YES instance $p_x = 1$, its perfect eigenspace has dimension six; in a NO instance $p_x \leq 1/3$, that dimension is one. Combining the circuit history with the transfer theorem produces nested weighted clique complexes with

$$\beta_d(X_{\text{in}}) = 8, \quad \text{rank}[H_d(X_{\text{in}}) \rightarrow H_d(X_{\text{out}})] = \begin{cases} 6, & x \in L, \\ 1, & x \notin L. \end{cases}$$

The normalized values are $3/4$ and $1/8$. The hardness source is the exact gate-dependent class $\text{BQP}_1^{G_2}$, with $G_2 = \{X, CX, CCX, H \otimes H\}$. Rudolph’s exact simulation results extend the statement separately to the fixed cyclotomic gate families covered there [7]; they do not yield a gate-independent theorem.

Finally, we use Hayakawa’s common-copy blow-up [3] with identical labeled copy blocks at every filtration level. The symmetric sector is an isometric copy of the weighted complex, up to a common Laplacian scale, and the asymmetric sector has a separate positive floor. The common labeling makes the isometries commute with inclusion. Consequently, endpoint Betti numbers, persistent rank, the normalized ratio, and whole-kernel gaps transfer to unweighted clique filtrations. This step is a sourced corollary rather than a new unweighting mechanism.

Our contributions are:

1. a finite-certificate all-chain leakage theorem for degenerate logical kernels, with a gadget scale independent of the logical gap;
2. exact kernel multiplicity and a whole-positive-spectrum gap obtained from the all-chain estimate and exact filling;
3. a functorial quotient realization for nested projector term sets, which computes the induced homology rank without compatible harmonic representatives;
4. a fixed-eight, gate-dependent BQP_1 -hardness theorem for true normalized persistence, together with a common-copy unweighted corollary.

The scope is intentionally narrow. Exact multiplicity, particular whole-kernel gaps, the normalized target, the individual few-term gadgets, and single-level unweighting all have close precedents. Our proof contribution is the corrected finite-certificate control of arbitrary geometric chains and its integration with a natural quotient filtration. We make no claim of ordinary BQP , unrestricted SDQC_1 , arbitrary-gate BQP_1 , or complex-phase hardness. We also do not claim that the inverse-polynomial gap is useful for near-term computation.

The rest of the paper defines the geometric and logical models, proves the transfer theorem, derives quotient naturality, instantiates the circuit and fixed palette, and transfers the construction to unweighted graphs. The appendices give the all-chain inequalities and the exact source calculations in expanded form.

2 Preliminaries and problem statement

All chain spaces are finite-dimensional over $\mathbb{C}$. A vertex-weighted simplicial complex assigns a positive weight $w(v)$ to every vertex and the product weight $w(\sigma) = \prod_{v \in \sigma} w(v)$ to a simplex. We use normalized simplex coordinates, so weighted boundary maps are diagonal rescalings of ordinary incidence matrices. For a complex X , let

$$\partial_{X,d} : C_d(X) \longrightarrow C_{d-1}(X), \quad \Delta_{X,d} = \partial_{X,d}^* \partial_{X,d} + \partial_{X,d+1} \partial_{X,d+1}^*.$$

Write $Z_d(X) = \ker \partial_{X,d}$, $B_d(X) = \text{ran } \partial_{X,d+1}$, and $H_d(X) = Z_d(X)/B_d(X)$. Finite-dimensional Hodge theory identifies $H_d(X)$ isometrically with $\ker \Delta_{X,d}$, and hence $\beta_d(X) = \dim \ker \Delta_{X,d}$.

For an inclusion $J : X_1 \hookrightarrow X_2$, the degree- d persistent Betti number is

$$\beta_d(X_1 \rightarrow X_2) := \text{rank}[H_d(X_1) \longrightarrow H_d(X_2)].$$

When $\beta_d(X_1) > 0$, define the true normalized persistence ratio

$$q_d(X_1, X_2) := \frac{\beta_d(X_1 \rightarrow X_2)}{\beta_d(X_1)}. \quad (2)$$

This denominator is the initial Betti number. It is unrelated to the number of degree- d simplices. We require an inverse-polynomial lower bound on every positive eigenvalue of both endpoint Laplacians. Thus q_d is an exact-kernel statistic rather than a count of eigenvalues below a nonzero energy cutoff.

Definition 2.1 (Gapped true normalized persistence). *An instance consists of polynomial-size graph descriptions of nested clique complexes $X_1 \subseteq X_2$, a degree d , a promised lower bound $E \geq 1/\text{poly}(n)$ on the positive spectrum of both $\Delta_{X_1,d}$ and $\Delta_{X_2,d}$, and the promise $\beta_d(X_1) > 0$. The task is to approximate [equation \(2\)](#) to a stated additive accuracy.*

The construction starts from a top-dimensional register complex R . Top-dimensional means $C_{d+1}(R) = 0$, and we set

$$V := Z_d(R) \subseteq C_d(R).$$

The register is a tensor join of normalized disjoint-edge cycles, so its logical embedding into V is explicit and isometric. A finite collection of rank-one projectors Π_j acts on V . For a term set A , define

$$H_A := \sum_{j \in A} \Pi_j, \quad K_A := \ker H_A, \quad W_A := \sum_{j \in A} \text{ran } \Pi_j. \quad (3)$$

Because H_A is positive semidefinite, $W_A = \text{ran } H_A = K_A^\perp$ inside V , and $\dim(V/W_A) = \dim K_A$.

Each Π_j is implemented by attaching a weighted clique gadget Y_j along the register. Gadget vertices have weight $\lambda \in (0, 1]$, register vertices have weight one, and private interiors of distinct gadgets share no simplex. The resulting clique complex is denoted X_A , and its full degree- d geometric Laplacian is Δ_A . The distinction between H_A and Δ_A is central: H_A acts on the logical cycle register, while Δ_A acts on every geometric d -chain.

We assume a known logical gap

$$H_A \succeq g_A(P_V - P_{K_A}), \quad 0 < g_A \leq 1, \quad (4)$$

where projectors are extended by zero outside V . The goal is to transfer [equation \(4\)](#) to the full spectrum of Δ_A , preserving the entire zero-space multiplicity.

For the complexity application, an exact circuit U has a maximally mixed logical input, computational-basis clean work qubits, and one computational-basis output measurement P_{acc} . If J inserts the fixed clean string, its compressed acceptance operator is

$$M = J^* U^* P_{\text{acc}} U J, \quad 0 \preceq M \preceq I. \quad (5)$$

The perfect accepting space is $S = \ker(I - M)$. The final history Hamiltonian will have kernel isomorphic to S , while the initial history Hamiltonian has a known kernel isomorphic to the entire input space.

3 Finite-certificate transfer to the complete geometric kernel

We state the local hypotheses in a form that can be certified once for a fixed gadget palette. The proof never appeals to convergence of a spectral projector in a chain space whose dimension grows with the simulated circuit.

3.1 Local certificates

Consider one active gadget before padding. Let R_a be its weight-one active register, let $V_a = Z_{d_a}(R_a)$, and give each new vertex weight λ . In normalized chain coordinates define the degree- d_a differential pair

$$D_a(\lambda) = \begin{bmatrix} \partial_{d_a}(\lambda) \\ \partial_{d_a+1}(\lambda)^* \end{bmatrix} = D_{a,0} + \lambda D_{a,1}.$$

The value at $\lambda = 0$ is a matrix limit; no singular change of basis at zero is used.

Assumption 3.1 (Finite local certificate). *For every member a of a fixed finite palette, there is a private-coordinate subspace Q_a , an output projector R_a^{out} , and a rank-one projector Π_a on V_a such that:*

(i) $B_{d_a}(Y_a) \cap V_a = \text{ran } \Pi_a$ for every positive λ ;

(ii) $\ker D_{a,0} = V_a \oplus Q_a$, orthogonally;

(iii) with $T_a(\lambda) = R_a^{\text{out}} D_a(\lambda)$,

$$T_a(\lambda) = \lambda T_{a,1}, \quad T_{a,1} V_a = 0, \quad \|T_{a,1} q\| \geq b \|q\| \quad (q \in Q_a);$$

(iv) all retained output coordinates in R_a^{out} contain a private vertex of the gadget.

The constant $b > 0$, all matrix norms, and the positive spectral floors at $\lambda = 1$ are uniform over the palette.

These conditions are finite linear algebra. Exact filling chains certify (i). Kernel vectors and modular rank certificates certify (ii). A Gram determinant or rational singular-value lower bound certifies (iii). The private support in (iv) prevents cancellation between different gadgets.

The elementary diagonal weight scaling gives a coarse local gap. If all active degrees satisfy $d_a = 2m_a - 1$ with $m_a \leq m$, then

$$\gamma_+(\Delta_{Y_a, d_a}(\lambda)) \geq c \lambda^{4m+2} = c \lambda^\kappa, \quad \kappa := 4m + 2. \quad (6)$$

Indeed, writing W_k for the diagonal matrix of simplex weights gives $\partial_{w,k} = W_{k-1}^{-1} \partial_{1,k} W_k$. Since $\sigma_{\min}(W_k) \geq \lambda^{k+1}$, every nonzero singular value is bounded below by a fixed unweighted singular value times λ^{k+1} . The finite palette turns the remaining constants into a common $c > 0$.

3.2 Padding and interfaces

An active gadget is joined with the unused register factors. We use reduced chain complexes, including degree -1 , and decompose the target chain degree over all join bidegrees. The unused register has reduced homology only in its top degree and a constant positive gap elsewhere. The join identity

$$\Delta_{Y_a * R_{\text{out}}} = \Delta_{Y_a} \otimes I + I \otimes \Delta_{R_{\text{out}}}$$

then shows that the padded zero-weight kernel is precisely

$$V \oplus (Q_a \otimes \mathcal{H}_{\text{out}}).$$

The private output projector is tensored with the outside harmonic projector. Outside differentials vanish on this subspace, so the fixed active estimate survives padding with constants independent of the outside dimension.

This harmonic restriction is necessary. If one instead defines the padded bulk as every coordinate containing a local central vertex, deleting a weight-one outside vertex remains in that coordinate bulk with coefficient one. The corresponding norm is not $O(\lambda)$. Thus the proof controls local bulk tensored with outside harmonics and sends every nonharmonic outside factor to the high-energy sector.

Attach t_A padded gadgets to the common register R . Distinct gadgets have disjoint private chain blocks. Let L_j map the private input coordinates of gadget j to the shared register output. We assume the fixed-interface estimate

$$L_j L_j^* \preceq N_j \lambda^2 I, \quad N_j \leq N_*. \quad (7)$$

The inputs of the L_j are orthogonal, although their outputs lie in the same register. Hence for the horizontal concatenation $L = [L_1 \ \cdots \ L_{t_A}]$,

$$L L^* = \sum_j L_j L_j^* \preceq N_* t_A \lambda^2 I.$$

No output-orthogonality assumption is present here. Orthogonality is used later only for the retained private maps T_j .

3.3 All-chain theorem

Theorem 3.2 (Finite-certificate degenerate-kernel transfer). *Fix a palette satisfying [assumption 3.1](#) and [equations \(6\)](#) and [\(7\)](#), use outside-harmonic padding in every bidegree, and attach gadgets with independent private interiors. Let $\mathcal{A}$ be any finite nested family of term sets, $t_{\max} = \max_{A \in \mathcal{A}} |A|$, and suppose [equation \(4\)](#) holds for every $A \in \mathcal{A}$.*

There are palette constants $c_0, C > 0$ such that, whenever

$$0 < \lambda \leq c_0 / t_{\max},$$

every unit chain $x \in C_d(X_A)$ satisfies

$$\|(I - P_{K_A})x\|^2 \leq C \left[t_A \lambda^2 + \frac{\langle x, \Delta_A x \rangle}{g_A \lambda^\kappa} \right]. \quad (8)$$

Here P_{K_A} denotes the projector onto the embedded logical kernel, extended by zero on all other chains.

For every $0 < \eta < 1$, choose additionally

$$\lambda \leq c_1 \min\{t_{\max}^{-1}, \eta / \sqrt{t_{\max}}\}, \quad E_A \leq c_2 \eta^2 g_A \lambda^\kappa. \quad (9)$$

Then

$$\dim \ker \Delta_A = \dim K_A, \quad \text{spec}(\Delta_A) \cap (0, E_A) = \emptyset. \quad (10)$$

The conclusion includes $K_A = 0$.

The proof has two independent parts. First, zero-weight coercivity, the private projected pair, and interface absorption give the purely geometric leakage estimate

$$\|(I - P_V)x\|^2 \leq C [t_A \lambda^2 + (t_A + \lambda^{-2}) \langle x, \Delta_A x \rangle]. \quad (11)$$

Second, exact filling and the local positive floor imply

$$\Delta_A^\uparrow \succeq c\lambda^\kappa H_A^{\text{emb}}. \quad (12)$$

To justify [equation \(12\)](#), note explicitly that $\text{ran } \Pi_j \subseteq B_d(Y_j)$, boundary chains are orthogonal to the local harmonic kernel, and the up-Laplacian has gap at least [equation \(6\)](#) on its range. Since R is top-dimensional and private interiors are independent, all degree- $(d+1)$ columns partition by gadget. Summing the local inequalities gives [equation \(12\)](#).

Combining [equations \(4\), \(11\) and \(12\)](#) proves [equation \(8\)](#). The full calculation appears in [appendix A](#). The estimate is uniform over every geometric chain, so it applies to every vector in a low-energy spectral subspace, including arbitrary linear combinations of eigenvectors.

Remark 3.3 (Relation to the positive-definite argument). *Suppose one already had an arbitrary-chain inequality*

$$1 - \langle x, P_V x \rangle + \langle x, H_A^{\text{emb}} x \rangle \leq R(e), \quad e = \langle x, \Delta_A x \rangle.$$

Then [equation \(4\)](#) immediately gives

$$\|(I - P_{K_A})x\|^2 \leq R(e)/g_A.$$

Thus the substitution of a degenerate logical gap and the later min-max closure are elementary once a correct uniform all-chain estimate exists. The contribution of [theorem 3.2](#) is the finite-certificate proof of that estimate, including the padded-sector repair and the independence of λ from g_A .

For $m \leq 6$ in the explicit history palette, $\kappa = 26$. Taking $\lambda = \Theta(\eta/t_{\max})$ in [equation \(9\)](#) yields the conservative scale

$$E_A = \Omega\left(\frac{\eta^{28} g_A}{t_{\max}^{26}}\right). \quad (13)$$

The theorem uses only a polynomial lower bound; no claim of an optimal exponent is intended.

4 Filtered quotient naturality

The transfer theorem determines the dimension of every endpoint kernel. Exact filling determines the maps between those kernels at the homology level. This separation is useful: spectral concentration proves that there are no extra classes, while the quotient model computes persistence without comparing harmonic representatives.

Lemma 4.1 (Independent filling). *For every term set A ,*

$$B_d(X_A) \cap V = W_A = \sum_{j \in A} \text{ran } \Pi_j. \quad (14)$$

Proof. Every vector in $\text{ran } \Pi_j$ has a filling in gadget j , so $W_A \subseteq B_d(X_A) \cap V$. Conversely, write a degree- $(d+1)$ chain as a sum of its gadget components. If its total boundary lies in V , then its boundary has zero coordinate on every private d -simplex. Private supports of distinct gadgets are disjoint, so this vanishing holds separately for each gadget component. The remaining register boundary of component j lies in $B_d(Y_j) \cap V = \text{ran } \Pi_j$. Summing over j proves the reverse inclusion. $\square$

No relative-acyclicity hypothesis appears in [lemma 4.1](#). Extra geometric homology outside V is excluded later by the all-chain concentration theorem, not by the boundary decomposition alone.

Theorem 4.2 (Natural quotient realization). *Under the hypotheses and parameter choice of [theorem 3.2](#), the map*

$$q_A : V/W_A \longrightarrow H_d(X_A), \quad q_A(v + W_A) = [v], \quad (15)$$

is an isomorphism. For $A \subseteq B$, these isomorphisms make the diagram

$$\begin{array}{ccc} V/W_A & \xrightarrow{\rho_{AB}} & V/W_B \\ q_A \downarrow \cong & & \downarrow \cong q_B \\ H_d(X_A) & \xrightarrow{(\iota_{AB})_*} & H_d(X_B) \end{array}$$

commute, where ρ_{AB} is the natural quotient map. Consequently,

$$\text{rank}(\iota_{AB})_* = \dim(V/W_B) = \dim K_B. \quad (16)$$

Proof. [Equation \(14\)](#) makes q_A well-defined and injective. Its domain has dimension

$$\dim(V/W_A) = \dim K_A.$$

By Hodge theory and [equation \(10\)](#), the codomain has dimension $\beta_d(X_A) = \dim \ker \Delta_A = \dim K_A$. Thus q_A is surjective as well.

If $A \subseteq B$, both paths send $v + W_A$ to the class of the same register cycle v in X_B . The diagram therefore commutes. Since $W_A \subseteq W_B$, ρ_{AB} is onto, and its rank equals the dimension of V/W_B . Finally, $W_B = \text{ran } H_B = K_B^\perp$ inside V , which proves [equation \(16\)](#). $\square$

For a chain $A_0 \subseteq A_1 \subseteq \dots \subseteq A_s$, every arrow in degree d is an epimorphism. The constructed barcode therefore has deaths but no new births in this degree. More generally, for any $i \leq j$,

$$\text{rank}[H_d(X_{A_i}) \rightarrow H_d(X_{A_j})] = \dim K_{A_j}.$$

The result concerns the algebraic homology maps. Harmonic subspaces at different filtration levels need not be related by a literal common isometry; their Hodge representatives can rotate while the quotient maps remain exact.

Taking A_{in} to be the clock, propagation, and clean-input terms of a history Hamiltonian, and taking A_{out} to add the final rejection term, gives

$$q_d(X_{\text{in}}, X_{\text{out}}) = \frac{\dim K_{\text{out}}}{\dim K_{\text{in}}}. \quad (17)$$

This identity is the bridge from exact logical perfect-space fractions to true normalized persistence.

5 Exact histories, a fixed clique palette, and hardness

We now instantiate the logical projector sums used in [theorem 3.2](#). The source is an exact real-gate perfect-completeness computation rather than an arbitrary trace-threshold computation.

5.1 A rational history for G_2

Let

$$G_2 = \{X, CX, CCX, H \otimes H\}.$$

For each $H \otimes H$ transition, write $K = \sqrt{2}H$. On a unary clock bit c and one work bit define the normalized forbidden vectors

$$\begin{aligned} u_0 &= (|00\rangle + |01\rangle - |10\rangle)/\sqrt{3}, & u_1 &= (|00\rangle - |01\rangle - |11\rangle)/\sqrt{3}, \\ d_0 &= (|00\rangle - |10\rangle - |11\rangle)/\sqrt{3}, & d_1 &= (|01\rangle - |10\rangle + |11\rangle)/\sqrt{3}. \end{aligned}$$

The two vectors in each pair are orthonormal. The corresponding projector sums enforce respectively

$$f_{t+1} = \sqrt{2}Hf_t, \quad f_{t+2} = Hf_{t+1}/\sqrt{2}.$$

Apply the gain edge to the first work bit and the consecutive loss edge to the second. Their composite action is exactly $H \otimes H$, and every rank-one projector has rational matrix entries. This gain/loss construction and its underlying three-term states are credited to Rudolph [7].

Transitions for X, CX , and CCX use the usual two-time vectors

$$(|0, z\rangle - |1, Uz\rangle)/\sqrt{2}.$$

For a fixed computational-basis string z , such a vector factors into basis guards and either a one-qubit difference or a two-qubit difference state. Clock, input, and output penalties are basis projectors. Consequently every local term belongs to a fixed finite collection of rank-one real projectors with total support locality at most six.

The clique implementation uses only explicit members of this collection. The base palette consists of a basis cone, one one-qubit difference gadget, the two two-qubit difference states related by a register relabeling, and four Hadamard three-term states related by exact petal relabelings. Computational-basis tensor factors are imposed by a selected-cycle guard. For each resulting gadget, exact integer filling, the zero-weight kernel, and the projected private differential admit finite certificates of the form in [assumption 3.1](#). This avoids any appeal to a general construction for arbitrary integer states.

5.2 The two history kernels and their gaps

Let L be the number of legal clock positions after splitting each Hadamard-pair transition. Put $s_t = \sqrt{2}$ immediately after a gain edge and $s_t = 1$ at every other legal position, and let V_t be the unitary prefix after removing these scalar factors. Then

$$\mathcal{V}v = Z^{-1/2} \sum_{t=0}^{L-1} s_t |t\rangle V_t v, \quad Z = \sum_t s_t^2, \quad L \leq Z \leq 2L. \quad (18)$$

For the valid clean-input space C , $\mathcal{V} : C \rightarrow \mathcal{K}_{\text{in}}$ is an isometry onto the complete zero space of the initial clock, propagation, and input Hamiltonian. Every zero-energy legal state is constant in prefix-rotated coordinates, and the input anchor forces that constant vector into C ; illegal clocks receive a positive clock penalty. Hence

$$\dim \mathcal{K}_{\text{in}} = D := \dim C, \quad g_{\text{in}} \geq \frac{1}{8L^2}. \quad (19)$$

The gap follows from a weighted path Poincaré inequality with vertex weights $s_t^2 \in [1, 2]$, applied separately to the clean mean-zero sector and the dirty anchored sector.

Let M be the compressed acceptance operator from [equation \(5\)](#), and let $S = \ker(I - M)$. Add the final rejection projector. Its compression to the initial history space is

$$\mathcal{V}^* H_{\text{rej}} \mathcal{V} = (I - M)/Z. \quad (20)$$

Therefore positivity gives

$$\ker H_{\text{out}} = \mathcal{V}S.$$

Assume the off-perfect promise $M|_{S^\perp} \preceq I/3$. [Equation \(20\)](#) gives an angle penalty at least $1/(3L)$ on $\mathcal{K}_{\text{in}} \ominus \mathcal{V}S$. The sharp two-positive-operator bound then yields

$$g_{\text{out}} \geq \frac{1}{3L(8L^2 + 1)} \geq \frac{1}{27L^3}. \quad (21)$$

This argument also covers $S = 0$, in which case [equation \(21\)](#) is a lower bound on the entire spectrum.

Combining [theorems 3.2](#) and [4.2](#) and [equations \(19\)](#) and [\(21\)](#) gives weighted clique complexes $X_{\text{in}} \subseteq X_{\text{out}}$ with

$$\beta_d(X_{\text{in}}) = D, \quad \beta_d(X_{\text{out}}) = \beta_d(X_{\text{in}} \rightarrow X_{\text{out}}) = \dim S, \quad (22)$$

and inverse-polynomial gaps above both complete geometric kernels.

5.3 An eight-label perfect-space source

Let Q_x be a uniform exact G_2 circuit on a fixed all-zero input, with one computational-basis output bit and

$$x \in L \implies p_x = 1, \quad x \notin L \implies p_x \leq \frac{1}{3}. \quad (23)$$

Add a maximally mixed label $z \in \{0, 1\}^3$, run Q_x unconditionally, and accept according to the reversible Boolean predicate

$$A(z, q) = [z = 000] \vee (q \wedge [z \in \{001, 010, 011, 100, 101\}]).$$

The predicate uses only X, CX, CCX , with scratch uncomputed before the final measurement. It does not control Q_x , so it introduces no controlled Hadamard. Because the label is preserved, the acceptance operator on the mixed label space is exactly

$$M_x = \text{diag}(1, p_x, p_x, p_x, p_x, p_x, 0, 0). \quad (24)$$

In a YES instance, six diagonal entries equal one. In a NO instance, only the unconditional block equals one and every other eigenvalue is at most $1/3$. Thus

$$\frac{\dim \ker(I - M_x)}{8} = \begin{cases} 3/4, & x \in L, \\ 1/8, & x \notin L. \end{cases} \quad (25)$$

Theorem 5.1 (Fixed-eight weighted hardness). *There is a polynomial reduction from exact $\text{BQP}_1^{G_2}$ verification to gapped true normalized persistence on nested weighted clique complexes. Every output satisfies*

$$\beta_d(X_{\text{in}}) = 8,$$

and the persistent rank is six in YES instances and one in NO instances. Both endpoint degree- d Laplacians have inverse-polynomial gaps above their complete kernels. Hence additive approximation of [equation \(2\)](#) with error less than $5/16$ is $\text{BQP}_1^{G_2}$ -hard; in particular, error $1/24$ and threshold $7/16$ suffice.

Proof. Apply [equation \(22\)](#) to [equation \(24\)](#). The denominator is eight and [equation \(25\)](#) gives the two target values. Their distance is $5/8$, so any additive error below half that distance separates them. The logical gaps [equations \(19\)](#) and [\(21\)](#) and the polynomial number of history terms make [equation \(13\)](#) inverse polynomial in the circuit size. $\square$

Rudolph proves exact simulation into G_2 for each fixed finite gate family over the covered power-of-two cyclotomic fields [7]. Composing that simulation with [theorem 5.1](#) gives hardness for each such gate family separately. Approximate universal compilation is insufficient because it need not preserve an eigenvalue-one space exactly.

The denominator in [theorem 5.1](#) is intentionally constant. Adding m ignored mixed bits tensors [equation \(24\)](#) with I_{2^m} , multiplying both multiplicities by 2^m and leaving the ratio unchanged. That closure is exact but padding-generated. We do not use it to suggest that the computation acts intrinsically on exponentially many distinct initial classes.

For comparison, an arbitrary normalized trace gap does not force a perfect-space fraction gap. If $M|_{S^\perp} \preceq rI$, $f = \dim S/D$, and $p = \text{Tr}(M)/D$, then

$$f \leq p \leq r + (1 - r)f.$$

The operators $I_{qD} \oplus 0$ and $I_{qD} \oplus \alpha I$ can have the same $f = q$ while satisfying opposite trace promises. Consequently [theorem 5.1](#) uses perfect completeness explicitly and does not imply unrestricted SDQC_1 hardness.

Finally, a circuit containing individual Hadamards can be represented using G_2 by adding one maximally mixed, unmeasured spectator a and replacing each H_j by $H_j \otimes H_a$. If the original circuit contains h Hadamards, then

$$U' = U \otimes H_a^h, \quad J' = J \otimes I_a, \quad P'_{\text{acc}} = P_{\text{acc}} \otimes I_a,$$

and, for either parity of h ,

$$M' = (J')^*(U')^*P'_{\text{acc}}U'J' = M \otimes I_a.$$

This exact spectator identity preserves all fractions and off-perfect gaps. It establishes no complex-phase extension.

6 A natural unweighted corollary

The weighted construction uses only weights 1 and a dyadic inverse-polynomial λ . Hayakawa's clique blow-up replaces a vertex v by a clique of f_v labeled copies and replaces each base edge by all edges between the corresponding blocks [3]. The block-symmetric sector reproduces the weighted Hodge Laplacian, while the asymmetric sector has an independent spectral floor.

Choose one integer F such that

$$f_v = Fw(v)^2 \in \mathbb{N}$$

for every vertex, and use the same labeled block for a shared vertex at every filtration level. For the two weights in our construction, choose dyadic λ and

$$F = \lambda^{-2}.$$

Then register vertices have F copies and gadget vertices have one copy. Since λ^{-1} is polynomial, the blown-up graph remains polynomial in size.

In degree k , let $U_{i,k}$ map a normalized base simplex to the normalized average of its one-copy-per-block lifts. With orientations fixed consistently across the filtration, Hayakawa's symmetric-sector identities read

$$\widehat{\partial}_{i,k} U_{i,k} = \sqrt{F} U_{i,k-1} \partial_{i,k}, \quad \widehat{\Delta}_{i,k} U_{i,k} = F U_{i,k} \Delta_{i,k}. \quad (26)$$

The common labeled blocks also give the exact inclusion identity

$$\widehat{J}_{ij} U_{i,d} = U_{j,d} J_{ij}. \quad (27)$$

There is no normalization factor in [equation \(27\)](#): inclusion preserves degree, weights, labels, and the set of copied simplices.

Let P_i and $\widehat{P}_i$ be the weighted and blown-up harmonic projectors. The asymmetric-sector lower bound excludes asymmetric harmonic vectors, so

$$\ker \widehat{\Delta}_{i,d} = U_{i,d} \ker \Delta_{i,d}, \quad \widehat{P}_i = U_{i,d} P_i U_{i,d}^*.$$

Combining this with [equation \(27\)](#) gives

$$\widehat{P}_j \widehat{J}_{ij} U_{i,d} = U_{j,d} P_j J_{ij}. \quad (28)$$

Thus every harmonic persistence map is unitarily equivalent to its weighted counterpart. Endpoint Betti numbers, singular values, induced ranks, and the ratio [equation \(2\)](#) are preserved exactly.

If E_i is a lower bound on the positive weighted spectrum, Hayakawa's asymmetric bound yields

$$\gamma_+(\widehat{\Delta}_{i,d}) \geq \min\{F E_i, \min_v f_v\}. \quad (29)$$

Here $\min_v f_v = 1$. Combining [equations \(13\)](#) and [\(29\)](#) gives the conservative unweighted floor

$$F E_i = \Omega\left(\frac{\eta^{26} g_i}{t_{\max}^{24}}\right)$$

until it reaches one.

To use a common vertex set, include every future gadget vertex at the initial level as an isolated vertex. The target degree in the reduction is positive, so these isolated vertices add no degree- d chain and do not alter the relevant Betti number or Laplacian. They acquire their prescribed edges when their gadget enters the filtration.

Corollary 6.1 (Unweighted true normalized persistence). *The reduction in [theorem 5.1](#) can be mapped in polynomial time to nested unweighted clique complexes with the same endpoint Betti numbers, persistent rank, and normalized values $3/4$ and $1/8$. Both unweighted endpoint Laplacians retain inverse-polynomial gaps above their complete kernels.*

This corollary uses Hayakawa's symmetric/asymmetric decomposition as a black box. The new observation needed for filtrations is [equation \(27\)](#), which follows from maintaining common labeled blocks. The unweighting mechanism itself is not claimed as a contribution.

7 Relation to prior work

Exact clique homology. Crichigno and Kohler give a parsimonious reduction from quantum satisfiability to clique homology: satisfying states correspond one-to-one with homology classes, yielding QMA_1 -hardness for existence and $\#\text{BQP}$ -hardness for exact counting [1]. This establishes that exact multiplicity alone is prior art. Their construction does not supply the inverse-polynomial Laplacian promise needed to identify exact zero modes robustly. Our target couples exact multiplicity with a gap above the complete kernel at two filtration endpoints and computes the inclusion rank between them.

Gapped clique homology. King and Kohler construct weighted clique gadgets and prove hardness for gapped clique homology [4]. Their many-gadget proof begins with arbitrary geometric chains, decomposes them into simulated and nonsimulated sectors, and applies positive definiteness of the logical Hamiltonian at the final coercivity step. Once a correct uniform arbitrary-chain master estimate is available, replacing positive definiteness by a gap above a degenerate logical kernel makes the final kernel and min-max argument short. We therefore do not claim the substitution itself as new.

Our technical difference lies earlier. In arXiv version 2, the proof of the padded coordinate-bulk estimate states that every relevant boundary coefficient contains a weight- λ vertex. A padded local bulk simplex can instead lose a weight-one outside-register vertex and remain in that coordinate bulk. The resulting term has coefficient one. We avoid this sector by projecting the fixed local private pair onto outside harmonics in every bidegree and placing nonharmonic outside factors in a separately gapped sector. We also use direct finite matrices rather than converting basiswise perturbation into a dimension-uniform projector estimate. This produces the $t\lambda^2$ structural error in equation (8). Our current source comparison is against arXiv version 2; whether the final SIAM text contains an equivalent repair must be checked before a priority claim is made.

Harmonic survival. Gyurik, Schmidhuber, King, Dunjko, and Hayakawa study whether a specified harmonic guide survives a filtration and prove gate-dependent BQP_1 -hardness for that problem [2]. Their construction also states a gap above the complete final harmonic space. The displayed proof invokes a lemma formulated on the simulated-register image while discussing an arbitrary low-energy geometric chain. We treat this as a proof-interface issue rather than a counterexample to their theorem. Our all-chain estimate is independent, uniform over arbitrary degenerate logical kernels, and used at both endpoints. The computational statistic is also different: one-guide survival does not return the fraction of the complete initial homology that survives.

Normalized persistence and persistent Laplacians. Lowe, Kim, Bondesan, and Hayakawa define the same initial-Betti-normalized ratio studied here and formulate hardness conjectures requiring exact multiplicity, endpoint gaps, and filtration compatibility [5]. They propose compatible harmonic isometries as a sufficient mechanism. We obtain the required numerical rank for a restricted family through the natural quotients V/W_A , without constructing a common harmonic-representative isometry. This is a restricted bypass, not a proof of their general compatibility conjecture or their unrestricted complexity conjecture.

Persistent Laplacians give a canonical spectral representation of persistent Betti numbers [6]. Their nullity equals the inclusion rank, and their operator decomposition already supplies standard domination relations between ordinary and persistent Laplacians. Our proof uses ordinary endpoint Laplacians to certify exact kernels and uses the quotient model to compute the induced map. We do not claim the general persistent-Laplacian formalism or its basic spectral inequalities.

Result type	Exact full multiplicity	Whole-kernel gap	Full-homology persistence ratio
Parsimonious clique homology [1]	yes	not supplied	no
Gapped clique homology [4]	NO-case focus	yes in stated promise	no
Guided harmonic survival [2]	not the target statistic	final construction	one guide
Normalized-persistence conjecture [5]	required	required	yes, conjectural clique step
This work	yes	both endpoints	yes, nested term sets

Table 1: The theorem package is distinguished by the conjunction of exact endpoint multiplicity, whole-kernel gaps, and a natural map on the complete initial homology. The table records problem scope rather than priority certification.

Exact gates and concrete gadgets. Rudolph develops exact gate-set simulations for perfect-completeness classes and supplies the gain/loss Hadamard factorization, concrete few-term states, and join closure used by our palette [7]. Those ingredients are credited. Our role is to restrict the reduction to a finite list, certify the exact local data needed by [assumption 3.1](#), and integrate the list with the all-chain and quotient arguments. No general integer-state clique-gadget theorem is assumed.

Unweighting. Hayakawa proves that common clique blow-ups reproduce a weighted Laplacian on a symmetric sector and give a positive floor on its orthogonal complement [3]. That result removes weights from gapped clique homology. Our filtration-level use keeps identical labeled blocks throughout the nested complexes, which gives [equation \(27\)](#). This naturality check is elementary, and [corollary 6.1](#) should be read as a direct application of Hayakawa’s theorem rather than an independent unweighting technique.

8 Limitations and conclusion

The result has four material boundaries. First, the transfer theorem is instantiated by a fixed finite real-gate palette. The local hypotheses are exact and checkable, but the construction does not cover arbitrary integer states or complex phases. Second, the complexity conclusion is gate dependent. It applies to $\text{BQP}_1^{G_2}$ and to the fixed cyclotomic gate families covered by an exact simulation theorem; it does not establish ordinary BQP, unrestricted SDQC_1 , or a gate-independent BQP_1 result.

Third, the cleanest theorem has the fixed denominator $\beta_d(X_{\text{in}}) = 8$. This is a strong structural restriction, but it also means that hardness is carried by deciding which of eight logical classes survive rather than by estimating a density over an intrinsically growing family of classes. Tensoring ignored mixed qubits makes the denominator grow while preserving the ratio, but that operation only replicates the same eight-label structure. An intrinsic growing-denominator theorem needs a different exact source whose perfect eigenspace fraction is hard without ancillary replication.

Fourth, the gap bound is designed for polynomial separation rather than numerical use. At locality six, the weighted estimate scales as $\Omega(\eta^{28}g/t^{26})$, and the history gap contributes $g = \Omega(L^{-3})$. The unweighted blow-up remains polynomial but can be large. These bounds support a complexity reduction and do not imply a near-term algorithmic advantage.

There is also a precise priority boundary. The qualitative degenerate-kernel closure is a short consequence of any valid King–Kohler-style arbitrary-chain master estimate. The present proof contribution is the finite-certificate replacement for two nonuniform padded steps and the resulting

logical-gap-independent choice of λ . Our comparison has checked arXiv version 2, while the detailed final SIAM text was unavailable during this draft. If that version already contains an equivalent fixed-local outside-harmonic repair with a quadratic $t\lambda^2$ remainder, the analytic novelty should be reduced accordingly. The theorem and its normalized-persistence application remain mathematically meaningful, but the paper’s positioning would shift toward synthesis and application.

Within these boundaries, the construction gives a complete route from degenerate logical kernels to true normalized persistence. A finite collection of local certificates controls every geometric chain. Exact filling supplies the intended zero modes, concentration rules out extras and opens a gap above the complete kernel, and quotient naturality turns endpoint control into an exact persistent rank. The fixed-eight source then yields separated normalized values, and a common-copy corollary removes the weights. The reusable output is the transfer criterion: future reductions can obtain gapped persistent multiplicities by verifying finite local zero-weight, private-coercivity, padding, filling, and interface conditions rather than repeating a growing-dimensional perturbation analysis.

AI Disclosure

OpenAI ChatGPT (GPT-5.6 Pro), OpenAI Codex, and Anthropic Claude were used to assist with research brainstorming, proof stress-testing, targeted literature discovery, manuscript drafting, and language editing. The authors independently checked the mathematical claims, proofs, source attributions, and final text and take full responsibility for the submission. No generative-AI system is an author or co-author.

References

- [1] Marcos Crichigno and Tamara Kohler. Clique homology is QMA_1 -hard. *Nature Communications*, 15:9846, 2024. doi: 10.1038/s41467-024-54118-z. URL <https://www.nature.com/articles/s41467-024-54118-z>.
- [2] Casper Gyurik, Alexander Schmidhuber, Robbie King, Vedran Dunjko, and Ryu Hayakawa. Provable quantum speedups for computing persistence in topological data analysis. *PRX Quantum*, 7:020361, 2026. doi: 10.1103/gvys-hl8h. URL <https://arxiv.org/abs/2410.21258v2>.
- [3] Ryu Hayakawa. Unweighted gapped clique homology is QMA_1 -complete, 2026. URL <https://arxiv.org/abs/2608.02726v1>.
- [4] Robbie King and Tamara Kohler. Gapped clique homology on weighted graphs is QMA_1 -hard and contained in QMA, 2024. URL <https://arxiv.org/abs/2311.17234v2>. Version 2.
- [5] Dominic Lowe, M. S. Kim, Roberto Bondesan, and Ryu Hayakawa. Complexity of normalized persistence problems for topological data analysis and local hamiltonians, 2026. URL <https://arxiv.org/abs/2607.03278v1>.
- [6] Facundo Mémoli, Zhengchao Wan, and Yusu Wang. Persistent laplacians: Properties, algorithms and implications, 2022. URL <https://arxiv.org/abs/2012.02808v3>. Version 3.
- [7] Dorian Rudolph. Towards a universal gateset for QMA_1 . In *51st International Symposium on Mathematical Foundations of Computer Science (MFCS 2026)*, 2026. doi: 10.4230/LIPIcs.MFCS.2026.98. URL <https://arxiv.org/abs/2411.02681v2>.

A Proof of the all-chain transfer theorem

This appendix proves [theorem 3.2](#) from the finite local hypotheses. All constants below depend only on the fixed palette.

A.1 Global decomposition and interface energy

Write the target chain space as the orthogonal coordinate sum

$$C_d(X_A) = C_d(R) \oplus \bigoplus_{j \in A} C_{d,j}^{\text{priv}}.$$

For a unit chain x , write

$$x = \omega_0 + \sum_j \omega_j, \quad x_j = \omega_0 + \omega_j, \quad U = \sum_j \|\omega_j\|^2, \quad e = \langle x, \Delta_A x \rangle.$$

Let $a = \partial_R \omega_0$, $y_j = L_j \omega_j$, $y = \sum_j y_j$, and $r = a + y$. The vector r is the shared-register component of the global down-boundary. Since $a + y_j = r - (y - y_j)$,

$$\sum_j \|a + y_j\|^2 \leq 2t_A \|r\|^2 + 2 \sum_j \|y - y_j\|^2.$$

The elementary identity

$$\sum_j \|y - y_j\|^2 = (t_A - 2) \|y\|^2 + \sum_j \|y_j\|^2$$

also holds for $t_A = 1$. Put

$$\rho = \lambda^{-2} \|[L_1 \cdots L_{t_A}]\|^2 \leq N_* t_A, \quad \chi = (t_A - 2)_+ \rho + N_* \leq C t_A^2.$$

Then [equation \(7\)](#) gives

$$\sum_j \|y - y_j\|^2 \leq \chi \lambda^2 U.$$

All nonregister down-boundary coordinates and all up-boundary coordinates are private and partition by gadget. We obtain

$$\sum_j \|D_j(\lambda) x_j\|^2 \leq 2t_A e + 2\chi \lambda^2 U. \quad (30)$$

A.2 Zero-weight and private-sector coercivity

Let $A_{j,0} = I - P_V - Q_j$ in the padded local chain space. The positive singular-value floor of $D_{j,0}$ on $\text{ran } A_{j,0}$, together with $D_j(\lambda) = D_{j,0} + \lambda D_{j,1}$, gives

$$\begin{aligned} S_A &:= \sum_j \|A_{j,0} x_j\|^2 \leq C \sum_j \|D_{j,0} x_j\|^2 \\ &\leq C \left(t_A e + \chi \lambda^2 U + \lambda^2 \sum_j \|x_j\|^2 \right) \\ &\leq C (t_A e + \chi \lambda^2 U + t_A \lambda^2). \end{aligned} \quad (31)$$

The final inequality uses $\sum_j \|x_j\|^2 = t_A \|\omega_0\|^2 + U \leq t_A$.

Because $T_j P_V = 0$,

$$T_j Q_j x_j = T_j x_j - T_j A_{j,0} x_j.$$

The lower bound on T_j restricted to Q_j , the estimate $\|T_j\| \leq C\lambda$, and orthogonality of the retained private output blocks imply

$$S_Q := \sum_j \|Q_j x_j\|^2 \leq C \left(\frac{e}{\lambda^2} + S_A \right). \quad (32)$$

The orthogonality here concerns the retained T_j outputs. The shared L_j outputs were handled by their horizontal operator norm and need not be orthogonal.

Let $N = \|(I - P_V)x\|^2$. Summing the orthogonal decompositions $P_V \oplus Q_j \oplus \text{ran } A_{j,0}$ gives

$$N \leq U + t_A \|(I - P_V)\omega_0\|^2 = S_A + S_Q.$$

Since $U \leq N$, substituting [equations \(31\)](#) and [\(32\)](#) and using $\chi \leq Ct_A^2$ yields

$$N \leq C [t_A \lambda^2 + (t_A + \lambda^{-2})e] + Ct_A^2 \lambda^2 N.$$

Choosing $\lambda \leq c_0/t_{\max}$ absorbs the last term and proves [equation \(11\)](#).

A.3 Logical coercivity

For each gadget, exact filling gives $\text{ran } \Pi_j \subseteq B_d(Y_j)$. The up-Laplacian has kernel $B_d(Y_j)^\perp$, and [equation \(6\)](#) lower-bounds its positive spectrum. Therefore, after the common embedding into X_A ,

$$\Delta_j^\uparrow \succeq c\lambda^\kappa \Pi_j.$$

The top-dimensional register has no up-columns, and every degree- $(d+1)$ simplex with a private vertex belongs to exactly one gadget. Hence the global up-Laplacian is the exact sum of the embedded local up-Laplacians, proving [equation \(12\)](#).

Since $K_A \subseteq V$, the squared distance decomposes as

$$\|(I - P_{K_A})x\|^2 = N + \langle x, (P_V - P_{K_A})x \rangle.$$

[Equations \(4\)](#) and [\(12\)](#) give

$$\langle x, (P_V - P_{K_A})x \rangle \leq \frac{e}{cg_A \lambda^\kappa}.$$

Together with [equation \(11\)](#),

$$\|(I - P_{K_A})x\|^2 \leq C \left[t_A \lambda^2 + (t_A + \lambda^{-2})e + \frac{e}{g_A \lambda^\kappa} \right].$$

For $\lambda \leq c_0/t_{\max}$, $g_A \leq 1$, and $\kappa \geq 2$, the last energy term dominates the middle energy terms up to a palette constant. This proves [equation \(8\)](#).

A.4 Exact multiplicity and the whole-positive-spectrum gap

Let $\mathcal{L}_A(E_A)$ be the span of all eigenvectors of Δ_A with eigenvalue below E_A . Every unit $x \in \mathcal{L}_A(E_A)$ obeys $e < E_A$. Choosing the constants in [equation \(9\)](#) makes the right-hand side of [equation \(8\)](#) strictly less than $\eta^2 < 1$. Therefore $P_{K_A}x \neq 0$ for every nonzero $x \in \mathcal{L}_A(E_A)$, so $P_{K_A}|_{\mathcal{L}_A(E_A)}$ is injective and

$$\dim \mathcal{L}_A(E_A) \leq \dim K_A. \quad (33)$$

Every $v \in V$ remains a cycle after attaching the gadgets. By [lemma 4.1](#), the map $V/W_A \rightarrow H_d(X_A)$ is injective. Its domain has dimension $\dim K_A$, so Hodge theory supplies at least $\dim K_A$ zero eigenvectors of Δ_A . These zero vectors belong to $\mathcal{L}_A(E_A)$. Combining this lower bound with [equation \(33\)](#) forces equality throughout: the geometric kernel has dimension $\dim K_A$, and no positive eigenvalue lies below E_A .

If $K_A = 0$, any unit vector in $\mathcal{L}_A(E_A)$ would have distance one from K_A , contradicting the strict $\eta < 1$ bound. Thus the same argument proves $\Delta_A \succeq E_A I$ in the zero-kernel case. No additional global relative-acyclicity assumption is required.

B History-gap and source calculations

B.1 Weighted path estimate

For a legal-clock state, write $f_t = s_t V_t \xi_t$, with the notation of [equation \(18\)](#). Its norm and initial-history energy are

$$N(\xi) = \sum_t w_t \|\xi_t\|^2, \quad E(\xi) = \sum_{t=1}^{L-1} c_t \|\xi_t - \xi_{t-1}\|^2 + \langle \xi_0, A_{\text{in}} \xi_0 \rangle,$$

where $w_t = s_t^2 \in [1, 2]$ and $c_t \in \{1/2, 2/3\}$. Split every ξ_t into the clean subspace $C = \ker A_{\text{in}}$ and its orthogonal complement.

For a state orthogonal to all valid histories, the clean component has weighted mean zero:

$$\sum_t w_t \xi_t^C = 0.$$

The weighted variance identity and path Cauchy–Schwarz inequality give

$$\sum_t w_t \|\xi_t^C\|^2 = \frac{1}{2 \sum_t w_t} \sum_{i,j} w_i w_j \|\xi_i^C - \xi_j^C\|^2 \leq 2L^2 E_C.$$

For the dirty component, set $a = \|\xi_0^{C^\perp}\|^2$ and $D_B = \sum_t \|\xi_t^{C^\perp} - \xi_{t-1}^{C^\perp}\|^2$. The bound

$$\|\xi_t^{C^\perp}\|^2 \leq 2a + 2(L-1)D_B$$

implies

$$\sum_t w_t \|\xi_t^{C^\perp}\|^2 \leq 8L^2 E_B.$$

Adding both sectors proves $N(\xi) \leq 8L^2 E(\xi)$ on the orthogonal complement of the complete initial kernel, establishing [equation \(19\)](#).

B.2 Adding the output projector

We record the two-positive-operator estimate used for [equation \(21\)](#). Let $A \succeq g(I - P)$, let R be an orthogonal projector, and let $\mathcal{S} = \ker A \cap \ker R$. Suppose the compression PRP is at least α on $\ker A \ominus \mathcal{S}$. Canonical two-projection blocks give

$$\gamma_+(A + R) \geq \frac{g + 1 - \sqrt{(g + 1)^2 - 4g\alpha}}{2} \geq \frac{g\alpha}{g + 1}. \quad (34)$$

For the history Hamiltonian, $g = 1/(8L^2)$ and $\alpha = (1 - r)/Z \geq 1/(3L)$ when $r \leq 1/3$. Substitution into [equation \(34\)](#) gives

$$\gamma_+(H_{\text{out}}) \geq \frac{1}{3L(8L^2 + 1)}.$$

The kernel identity follows directly from positivity:

$$\ker(H_{\text{in}} + H_{\text{rej}}) = \ker H_{\text{in}} \cap \ker H_{\text{rej}} = \mathcal{V} \ker(I - M).$$

B.3 Exact eight-label compression

Let U_x denote the fixed-input verifier and q its output bit. The label z is never changed. After computing and uncomputing the constant-size acceptance predicate, compression through all clean work registers has no matrix element between distinct labels. On $z = 000$, acceptance is the identity. On five labels it is the scalar $p_x = \|P_{\text{acc}}U_x|0 \cdots 0\rangle\|^2$, and on two labels it is zero. This proves [equation \(24\)](#) even for a coherent vector across label values.

The normalized trace is

$$\frac{\text{Tr } M_x}{8} = \frac{1 + 5p_x}{8}.$$

If $p_x = 1$, this equals $3/4$ and coincides with the perfect-space fraction. If $p_x \leq 1/3$, it is at most $1/3$, the perfect-space fraction is exactly $1/8$, and every off-perfect eigenvalue is at most $1/3$. Thus the construction simultaneously supplies the trace promise, the perfect-space separation, and the off-perfect ceiling needed by the output-gap proof.

B.4 Finite palette audit boundary

The local theorem requires exact data only for the following fixed families:

Logical vector type	Construction	Role
Basis vector	selected-petal cone	clock/input/output penalties
$ 0\rangle - 1\rangle$	explicit one-qubit gadget	unary difference
$ 00\rangle - 11\rangle$	explicit two-qubit gadget	ordinary propagation
$ 01\rangle - 10\rangle$	register relabeling	ordinary propagation
four three-term states	petal relabelings of one base gadget	gain/loss Hadamards
fixed basis factors	selected-cycle join guard	locality extension

For each base graph, the finite evidence consists of an exact filling chain, an exact kernel description at zero weight, and an injective projected private map. Register relabelings transport these identities exactly, and the selected-cycle guard tensors the chosen harmonic line while placing all other guard modes in a separately coercive sector. The analytic theorem uses these identities as hypotheses; it does not infer them from numerical spectra.

The arXiv-v2 King–Kohler comparison and the final-published-version comparison are distinct evidence states. The former motivates the corrected padding proof above. A final version of this manuscript must report the latter before making a priority claim about that repair.